

KEATON ZELAZNY

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Engineering and Computer Science

Expected May 2027

GPA: 3.94 / 4.00 | Dean's List | Global Scholars Program - Oakland, CA & London, UK (2023 - 2024)

Relevant Coursework: Fundamentals of Computer Science I & II; Embedded Design: Enabling Robotics; Computer Systems; Algorithms and Data; Fundamentals of Digital Design; Circuits and Signals: Biomedical Applications; Fundamentals of Networks; Discrete Structures; Logic and Computation; Linear Algebra and Differential Equations; Fundamentals of Electronics, Programming in C++, Computer Architecture and Organization, Introduction to Databases (in progress, Fall 2026)

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SystemVerilog, MATLAB, RISC-V Assembly, x86 Assembly, HTML

Hardware / EDA: SystemVerilog, Quartus Prime, Vivado Design Suite, PSpice, FPGA development (TUL PYNQ), analog circuit design, SolidWorks, AutoCAD

Software / ML: Gymnasium, PyBullet, Stable-Baselines3, TensorBoard, URDF, REST APIs, SQL

Tools: Git, Unix/Linux, GDB, Ignition, FactoryTalk, LaTeX

TECHNICAL PROJECTS

Project Jaeger — RL for Bipedal Locomotion

Fall 2025 – Present

- Python, Gymnasium, PyBullet — custom 6-DOF biped simulation; PPO policy trained as a residual correction on a physics-based PD controller.

Single-Cycle RISC-V Processor

Spring 2026

- SystemVerilog, Vivado, TUL PYNQ FPGA — implemented and synthesized a RISC-V-style datapath and control unit, validated on-board.

Custom Unix Shell

Spring 2026

- C, POSIX — command-line shell with process creation, I/O redirection, and pipelines built directly on fork, exec, and pipe.

ECG / EKG Analog Front End

Spring 2025

- Op-amp filter design, PSpice — cascaded amplification and filtering stages recovering a readable waveform from raw electrode input.

Six-Legged Walking Robot

Fall 2024

- Embedded C++ — gait and steering firmware coordinating servo timing across six legs, tuned against hardware behavior.

RFID Phone Lockbox

Spring 2024

- Microcontroller, RFID, servo actuation — access-controlled enclosure combining tag authentication with a servo-driven latch.

Recipe Book Application

Summer 2026

- Java — recipe storage and search application built around clean separation of data, logic, and interface layers.

Explicit Derivative Calculator

Winter 2024

- Python — symbolic calculator that differentiates explicit expressions by recursively applying rules down the parse tree.

WORK EXPERIENCE

Genezen

Lexington, MA

Automation Engineer Co-op

Jun 2025 – Dec 2025

- Replaced end-of-life sensor transmitters and rebuilt their real-time integration into an Ignition SCADA system, authenticating to the vendor REST API over OAuth2 and pulling live sensor data via Jython scripting.
- Designed parameterized sensor type definitions (UDTs) and operator display screens, wiring every instance into automatic SQL historian logging so new sensors required no per-unit configuration.
- Authored and independently executed validation test protocols for production equipment, producing the documentation record required for engineering and quality sign-off.

Integrated Mill Systems

Willoughby, OH

Fabricator

Summer 2023

- Assembled and bench-tested custom control panels for industrial automation systems.

Legacy Interiors

Chardon, OH

Painter (Seasonal, Contract)

Summers 2024 – 2026

- Completed residential interior and exterior painting projects with a small team of painters.

INTERESTS

Cooking, running, NBA basketball, video games, travel, reading, robotics, community engagement.